Naïl B. Khelifa

PhD student in Information Engineering
Department of Engineering and Trinity College, University of Cambridge

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Google Scholar GitHub LinkedIn

RESEARCH INTERESTS

Diffusion models, stochastic analysis, statistical learning and optimal transport.

EDUCATION

University of Cambridge 2025–2029
PhD in Information Engineering, Trinity College
Supervised by Prof. Ramji Venkataramanan and Prof. Richard E. Turner.

École Normale Supérieure Paris-Saclay 2024–2025
MSc in Mathematics, Vision and Learning (MVA)
Joint with the final-year track in data science, statistics and learning at ENSAE.
GPA 4.0/4.0.

ENSAE Paris, Institut Polytechnique de Paris 2021–2025
Diplôme d'ingénieur in Statistics and Applied Mathematics
Double degree with HEC Paris. GPA 4.0/4.0.

HEC Paris 2021–2025
Grande École programme in Management
Double degree with ENSAE Paris. GPA 3.54/4.0.

Sorbonne Université 2022–2023
Master's degree in Mathematics
Ranked 5th of 84.

Lycée Henri IV, Paris 2019–2021
Classe préparatoire aux grandes écoles (ECS)
Admitted to HEC Paris and ENSAE Paris. National entrance examination: 10th (written) and 44th (oral) of about 5,000 candidates.

PUBLICATIONS AND PREPRINTS

Diffusion Models Observe Only Gradients: A Geometric Perspective on Score Matching Errors NEURIPS 2026
Naïl Khelifa, Richard E. Turner and Ramji Venkataramanan
40th Annual Conference on Neural Information Processing Systems.

Recursively Trained Diffusion Models: Limiting Collapse Distribution and Spectral Characterization NEURIPS 2026
Naïl Khelifa, Richard E. Turner and Ramji Venkataramanan
40th Annual Conference on Neural Information Processing Systems. ***Spotlight.***

Error Propagation and Model Collapse in Diffusion Models: A Theoretical Study ICML 2026
Naïl Khelifa, Richard E. Turner and Ramji Venkataramanan
43rd International Conference on Machine Learning.

Enhanced Image Reconstruction and Convergent Regularisation Using Tweedie Scaling SSVM 2025
Naïl Khelifa, Carola-Bibiane Schönlieb and Ferdia Sherry
10th International Conference on Scale Space and Variational Methods in Computer Vision.

Cluster-driven Hierarchical Representation of Large Asset Universes for Portfolio Construction ICAIF 2024
Naïl Khelifa, Jérôme Allier and Mihai Cucuringu
5th ACM International Conference on AI in Finance. Oral presentation; Best Paper Award, honorary mention.

RESEARCH EXPERIENCE

École Polytechnique Jun–Oct 2025
Visiting student, Centre de Mathématiques Appliquées
Modelling graph dynamics via optimal transport autoencoders, with Prof. Rémi Flamary, Prof. Charlotte Laclau and Prof. Florence d'Alché-Buc.

University of Cambridge Jun–Oct 2024
Visiting student, Department of Applied Mathematics
Enhanced denoising and convergent regularisation using Tweedie scaling, with Dr Ferdia Sherry and Prof. Carola-Bibiane Schönlieb.

University of Oxford Nov 2023 – Dec 2024
Part-time research student (remote), Department of Statistics
Cluster-driven hierarchical representation of large asset universes for portfolio construction, with Prof. Mihai Cucuringu.

Technical University of Munich Jun–Aug 2023
Visiting student, Department of Financial Mathematics
Stable time series clustering, with Dr Florian Brück and Prof. Rudi Zagst.

TEACHING

MPhil thesis co-supervision Apr–Sep 2026
MPhil in Machine Learning and Machine Intelligence, Cambridge
"Learning Heavy-Tailed Distributions with Diffusion Models", co-supervised with Prof. Richard E. Turner.

Inference (Probabilistic Machine Learning) Jan–Apr 2026
Undergraduate supervisor, third year of the Engineering Tripos, Cambridge

Information Theory and Coding Oct 2025 – Jan 2026
Undergraduate supervisor, third year of the Engineering Tripos, Cambridge

TALKS AND POSTERS

Microsoft Research Mathematical foundations of AI workshop Oct 2026
Poster. Microsoft Research, Cambridge UK

G-Research Scholar Symposium Oct 2026
Talk. Isaac Newton Institute, Cambridge, UK

Princeton Machine Learning Theory Summer School Aug 2026
Poster. Operations Research and Financial Engineering, Princeton, USA

Duke Summer School in Stochastics, Dynamics, and ML Aug 2026
Poster. Department of Mathematics, Durham, USA

GenAI × ProbAI Hub Workshop: ML Methods and Algorithms Jun 2026
Poster. ProbAI Hub, London, UK

SCHOLARSHIPS AND AWARDS

NeurIPS 2026 Spotlight (top 3.6%) 2026
G-Research Trinity College Scholarship, Cambridge 2025–2029
Best Paper Award, honorary mention, ICAIF 2024

LANGUAGES AND SKILLS

Languages: French (native), English and German (proficient)
Programming: Python
Outside research: rowing, football, basketball, piano